



Away with the people!

Bye Bye Camera is an app which can remove all humans from a photo using an AI tool. <https://dgit.in/july19-20>



Raspberry Pi Hacker

A hacker has successfully penetrated NASA's network just by using a \$35 Raspberry Pi. <https://dgit.in/july19-21>

Credit: Wyss Institute at Harvard University

4D PRINTING

Forget 3D printing, the next big thing is here

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The process of 4D printing is a lot like 3D printing. You have a heated tip that adds a material layer by layer to form an object. Depending on the material used though, 4D printing can provide additional properties to the object. When exposed to heat, light, electricity, humidity, sound, pressure or other external stimuli, the objects change shape. Instead of static objects, the additive manufacturing allows for the creation of dynamic objects, that can be

used for a range of purposes. The inspiration for these new materials, called metamaterials or multimaterials, are plants. The leaves, tendrils and flowers of plants can change their shapes according to environmental inputs.

Researchers at MIT have 3D printed materials that can return to their original shape after being twisted or bent. On the application of heat, the objects spring back to their intended shapes, and are considered to have a "shape memory". Instead of layering the material, the researchers used a new method called microstereolithography. The required object is first designed in a CAD software, then sliced up. Each of the slices is converted to a bitmap, which is then projected on a liquid resin. The resin that has been exposed to light then solidifies. The process is then repeated

for the next layer, and the object is created using a stack of exposures. This approach allows the researchers to create much more fine-grained structures than traditional 3D printing. Nicholas X Fang, Associate Professor of mechanical engineering at MIT explained, "This is a very advanced 3-D printing method compared to traditional nozzle or ink-jet based printers. The method's main advantages are faster printing and better structural integrity." To demonstrate the applications of the technology, the researchers used the technique to create a soft gripper, that picked up screws when the ambient temperatures increased. The same gripper was also used to handle delicate objects such as fish eggs. Researchers at Rutgers University have also created materials that can be 4D printed into objects that use